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On the relationship between the cutting process, surface integrity and the corrosion resistance of OFHC copper

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Abstract

Enhancing the functional performance and life of high added value components through improving their surface integrity is of prime interest, especially when those components are used in critical applications like the nuclear industry. The aim of the present work consists in, first, investigating the impact of the cutting process parameters on the surface integrity in terms of mechanical and microstructure states, then, identifying the role of each surface integrity parameter in the corrosion resistance of the machined surface.

Experiences performed on OFHC copper have shown that the control of the process parameters enables the improvement of corrosion resistance and local electrochemical behaviours of the machined surface. Physical explanations of the impact of the process parameters on the surface integrity, then consequently on the corrosion resistance are discussed.

Keywords: corrosion; copper; surface integrity; machining.
